

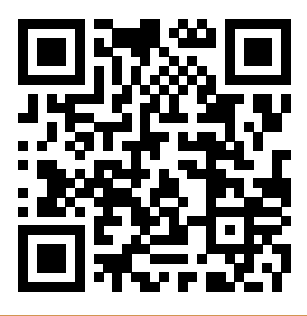
AGONPROJECT

An online platform for exploring approaches to formal argumentation

WEB PLATFORM • OPEN SOURCE

The whole zoo of formal argumentation, in one browser tab.

Formal argumentation is a rich but fragmented field. AgonProject gives students, researchers and practitioners a single, **visual** interface to construct, evaluate and compare its many formalisms — powered by the *TweetyProject* reasoning libraries.



TRY IT LIVE
agon.tweetyproject.org
github.com/aig-hagen/AgonProject • GPL-3.0

AgonProject

The platform to explore different approaches to formal argumentation.

Abstract Argumentation

The foundational model for the formal representation of argumentation via arguments and directed attacks.

OPEN EXAMPLE
meal_wine two_cycles
grounded_ideal initial_sets

NEW
Create new Generate random AF

Bipolar Argumentation

Relations between arguments can be either attacking or supporting.

OPEN EXAMPLE
meal_wine vacation_planning
murder_trial

NEW
Create new Generate random BAF

Dialectical Argumentation

Relations between arguments are modelled via propositional acceptance conditions.

OPEN EXAMPLE
meal_wine murder_trial

NEW
Create new Generate random ADF

Incomplete Argumentation

Distinguish between certain and uncertain arguments and attacks to represent incomplete information.

OPEN EXAMPLE
murder_trial

NEW
Create new Generate random IAF

Probabilistic Argumentation

Arguments and attacks can be assigned a probability value between 0 and 1 to quantify uncertainty.

OPEN EXAMPLE
murder_trial

NEW
Create new Generate random PAF

Argumentation with Collective Attacks

Extends abstract argumentation by allowing sets of arguments to collectively attack a target argument.

OPEN EXAMPLE
tandem_trip

NEW
Create new

One landing page — pick a formalism and start modelling, with curated examples and one-click random generation.

FEATURES

Build

Interactive graph editor with automatic layouts and a physics simulation.

Evaluate

Extension- and ranking-based semantics; enumeration, credulous / skeptical acceptance.

Export & share

LaTeX, SVG, TGF and ICCMA AF-format, plus share-URLs.

Generate

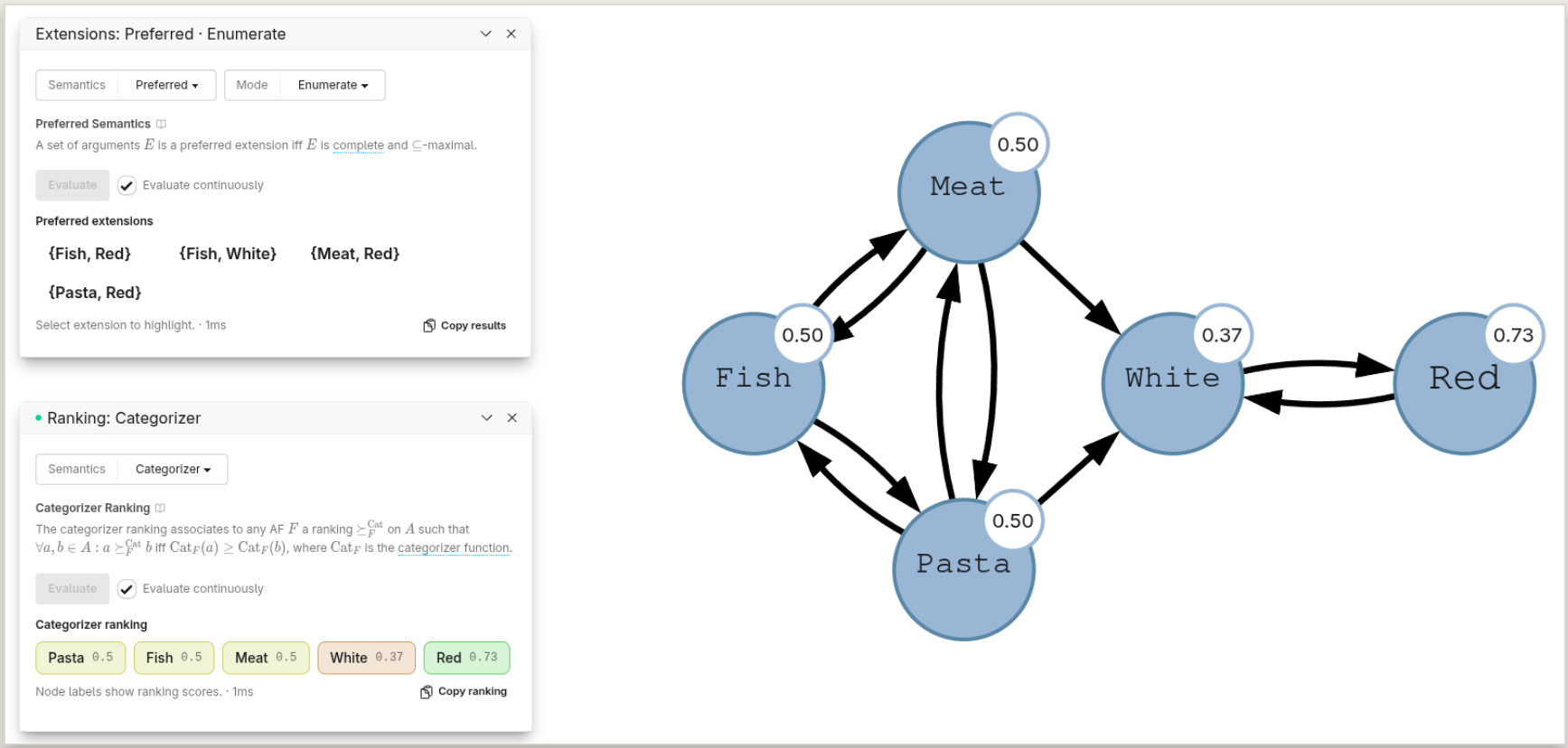
Random frameworks from eight configurable networkX generators.

Learn

10+ tutorials and a glossary with literature references.

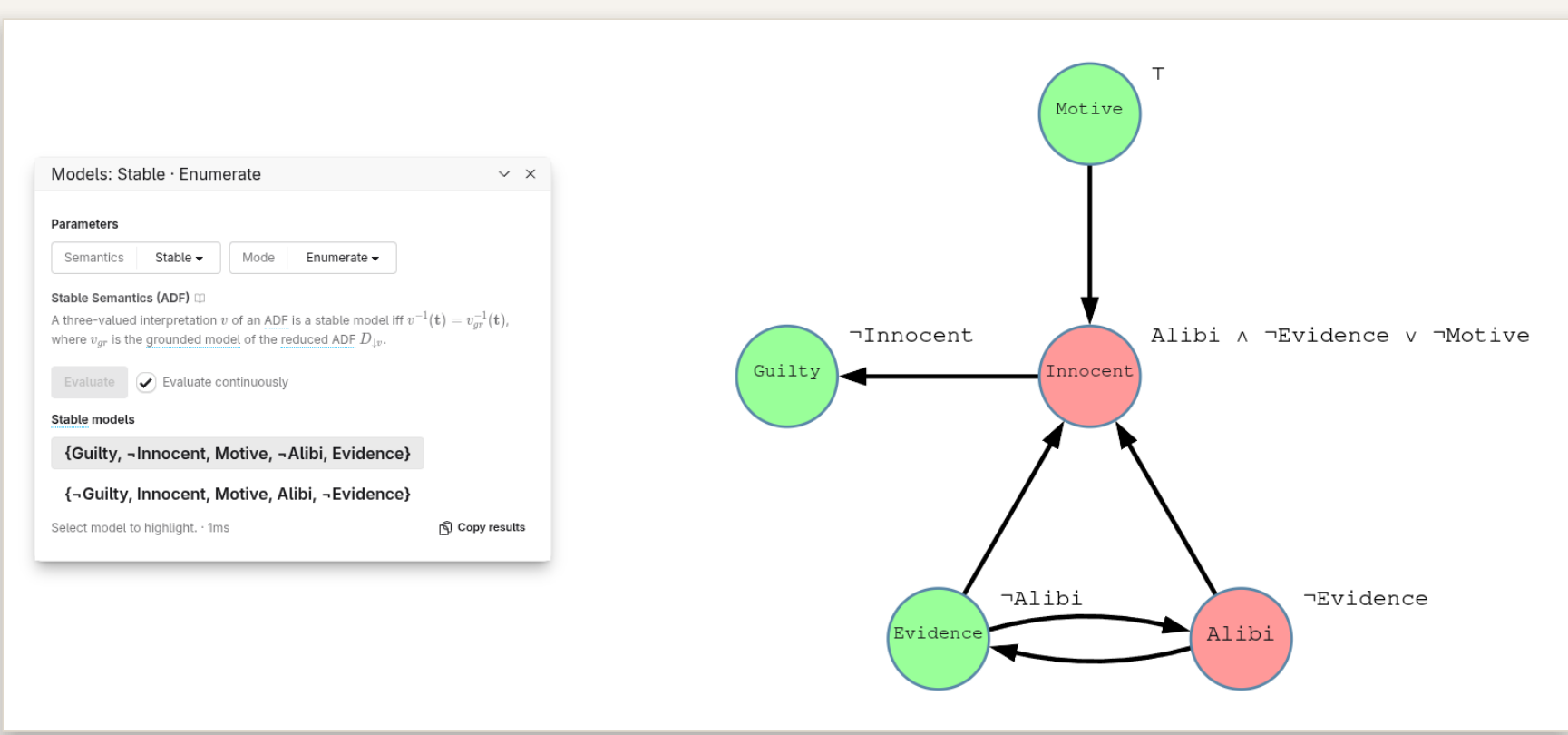
IN ACTION

Evaluate & rank



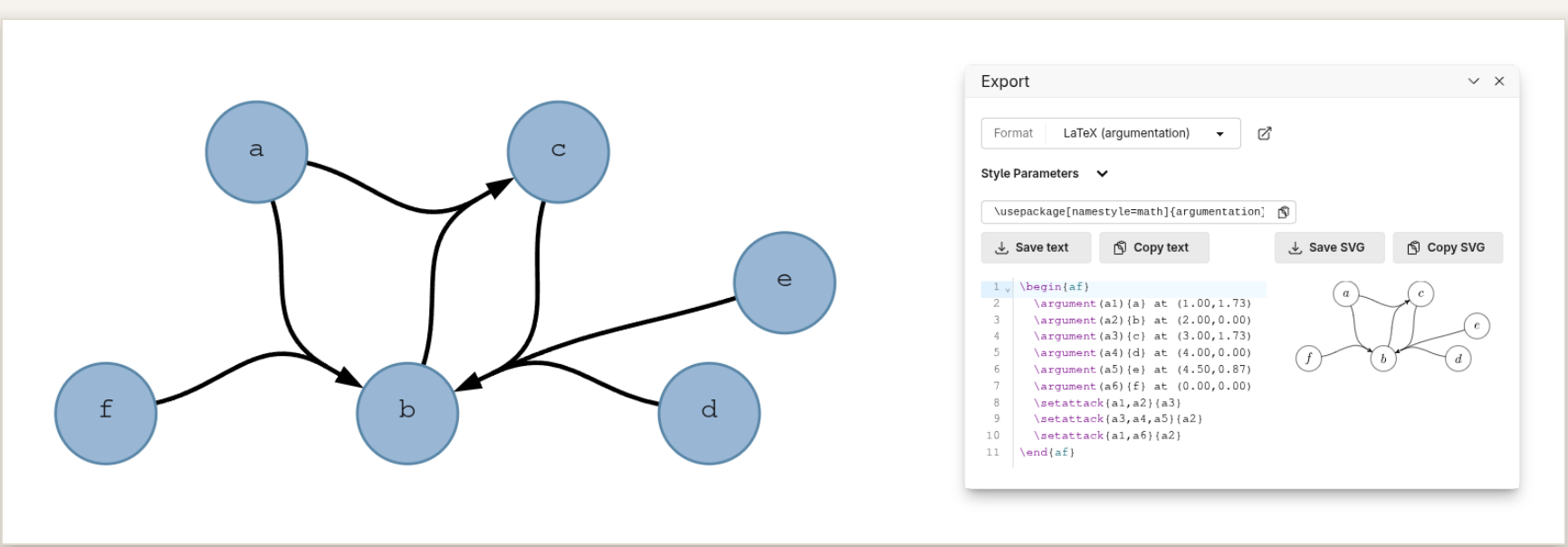
Preferred extensions enumerated, with the categoriser ranking annotated on each argument.

Acceptance conditions



ADFs attach propositional conditions to arguments; a stable model is highlighted in the graph.

Collective attacks & export



Set-attacks drawn as hyperedges; the export panel emits ready-to-use LaTeX with an SVG preview.

SUPPORTED SEMANTICS

Classical	conflict-free, admissible, complete, grounded, preferred, stable
Admissibility-based	ideal, semi-stable, eager, strong admissibility, initial sets, unchallenged
Non-admissible	naive, CF2, SCF2, stage, stage2, (strongly) undisputed
Rankings	burden, categoriser, counting, discussion, iterated graded defense, serialisability, ...

OUTLOOK

- more formalisms — ABA, weighted AFs, BSAFs, claim-augmented AFs
 - benchmark generators from the ICCMA ecosystem
 - explanation-based reasoning: *why* is an argument accepted?
- AgonProject is backend-agnostic and open to new formalisms and solvers.